Mauri de Souza Meneguzzo

Senior Software Engineer

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open source & community

- **Go** Contributor and codeowner in the Go project, covering the language runtime, compiler, assembler and ecosystem. Awarded the 2024 Google Open Source Peer Bonus. Driving 128-bit atomics support (targeting Go 1.28) and architecture-specific work for modern ARM64 cores.
- **LLVM** Contributor focused on compiler-rt/ThreadSanitizer (tsan) support for Go.
- **OpenTelemetry** Contributor focused on elasticsearchexporter and contrib. Nominated Top Contributor at KubeCon EU 2025.
- **Kubernetes** Contributor through bug fixes, documentation improvements and feature discussions.
- **Elastic** Open source contributor to Elastic Agent, Beats and related ingest tooling; most work is publicly available upstream.

work

2026 - ongoing

SYSTEMS ENGINEER, STREAMS/LOGSTASH *Elastic* **Remote / Full-time**

- Founding engineer on Streams, Elastic's next-generation unified data ingest and processing platform.
- Building the compiler and execution engine at the core of Streams, turning pipeline definitions into managed telemetry infrastructure.

2024 - 2026

SENIOR SOFTWARE ENGINEER, OPENTELEMETRY *Elastic* **Remote / Full-time**

- Member of the Ingest Data Plane team working on Elastic Agent, Beats and OpenTelemetry.
- Led performance work that reduced Elastic Agent memory usage by up to 50% and significantly lowered CPU consumption when operating in OpenTelemetry mode.
- Drove adoption of OpenTelemetry across Elastic products, including EDOT and Elastic Agent integrations.
- Upstream contributor to OpenTelemetry, particularly elasticsearchexporter and contrib.
- Participated in support rotations, resolving complex production issues, security vulnerabilities and customer escalations.

2022 - 2024

FULL STACK SOFTWARE ENGINEER *Meisterwerk* **Remote / Full-time / Germany**

- Designed and implemented Go-based HTTP and gRPC microservices serving production workloads.
- Improved application build times by 64% through caching and parallelization strategies.
- Led performance investigations using pprof and Delve, eliminating CPU and memory bottlenecks.
- Introduced automated security scanning and static analysis across CI/CD pipelines.
- Mentored engineers and established development best practices.

2021-2022

ASSOCIATE / FULL STACK SOFTWARE ENGINEER *Bucks App / Startup (finance)* **Remote / Full-time / USA**

- Developed secure Go and Node.js systems on AWS, integrated with financial solutions.
- Implemented cryptographically secure systems to ensure data integrity and confidentiality.

2016-2021

CTO / LEAD SOFTWARE ENGINEER *3CPLUS - SAAS (communications)* **Remote / Full-time**

- Led a team of 6 engineers; company achieved 120% MRR growth in 2020.
- Scaled real-time systems serving thousands of concurrent users and millions of events daily.
- Led migration to Kubernetes and GCP, enabling horizontal scalability across the platform.
- Designed distributed ingestion and event-processing systems powering analytics and reporting workloads.
- Improved ML inference workloads by 10x while reducing infrastructure requirements by 20x.
- Developed high-performance services in Go, Rust, C, PHP and Node.js, including Asterisk PABX modules.

2014-2015 **IT ASSISTANT, PRIVATE RANK** *6th Comms Bn of the 6th Div, Brazilian Army* **Onsite / Full-time**

- Linux server administration, network management, and IT support.
- Signal Corps, went to army reserve with an honorable mention.

education

2024-2025 **POSTGRADUATE SPECIALIZATION IN CYBERSECURITY AND DEFENSE** *Uninter, Brazil*

- Introduction to Software Simulation in Combat and Armored Warfare - *UFRGS, Brazilian Army*

2023-2024 **POSTGRADUATE SPECIALIZATION IN DATA SCIENCE AND AI** *Uninter, Brazil*

2017-2021 **BTECH IN IT WITH A FOCUS ON SYSTEMS ANALYSIS AND DESIGN** *Uniftec, Brazil*

astrophysics

Independent research on exoplanet detection in NASA TESS light curves.

- **transitgpu** GPU implementations of the Box Least Squares and Transit Least Squares transit searches in PyTorch and Triton, for CUDA and ROCm. github.com/mauri870/transitgpu
- **velociraptor** GPU drop-in for TRICERATOPS, which computes the probability that a transit signal is a false positive, using an exact transit light curve. github.com/mauri870/velociraptor

papers

- M. de Souza Meneguzzo (2026). *Parallel Search for Alien Worlds: GPU-Accelerated BLS and TLS for Exoplanet Transit Detection*. Preprint.
- M. de Souza Meneguzzo (2026). *velociraptor: Exact Light Curves and GPU-Accelerated False-Positive Probabilities for TESS Planet Candidates*. Preprint.

languages

Portuguese (native)
English (fluent)

skills

Languages: Go, Rust, Python, TypeScript, PHP, C, Bash
Infrastructure: Kubernetes, Docker, AWS, GCP, Linux
Distributed Systems: Observability, OpenTelemetry, Event Processing, Microservices, High-Performance Systems
Tools: Git, GitLab CI/CD, Prometheus, Elasticsearch, Kibana